

FPGA-Controlled Focal Plane VCSEL Array for High-Speed Optical Wireless Communication

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Abstract—This paper reports an experimental demonstration of optical beam steering using a 2D VCSEL focal plane array (FPA) combined with a lens. The VCSEL array is driven by a field-programmable gate array (FPGA), enabling real-time electronic selection of individual emitters and corresponding data routing for dynamic beam direction control. The system achieves steering ranges of 10.1° horizontally and 15° vertically. Error-free data transmission at 10 Gbps is demonstrated over indoor free-space distances up to 4 m. These results verify the feasibility of a 4 × 4 VCSEL FPA for high-speed indoor OWC links, supporting dedicated multi-user connectivity while enhancing link security and privacy.

Index Terms—Focal-plane array, FPGA, multi-beam system, optical beam steering, optical wireless communication.

I. INTRODUCTION

THE rapid growth of data consumption, driven by the proliferation of Internet-of-Things devices, high-definition streaming, artificial intelligence workloads, and cloud-based services, is pushing traditional radio-frequency (RF) wireless networks toward their capacity limits [1]. The combination of limited available spectrum and strict regulatory constraints makes it increasingly difficult for RF systems to support dense, high-throughput wireless connectivity in indoor environments [2].

Optical wireless communication (OWC) offers a complementary solution by exploiting the unlicensed optical spectrum, which provides orders of magnitude more available bandwidth and inherently enhanced privacy. Because light does not penetrate walls, OWC links are confined to a room, preventing inter-room interference and signal leakage. This spatial confinement allows OWC networks to offload high-capacity traffic from congested RF links while maintaining secure, localized data delivery [3].

Received 7 December 2025; revised 3 March 2026 and 4 May 2026; accepted 7 May 2026. Date of publication 14 May 2026; date of current version 16 July 2026. This work was supported in part by the Netherlands Organization for Scientific Research (NWO) through the National Perspective Program under Grant 12128, in part by 'FREE - Optical Wireless Super-highways: Free Photons, and in part by Growth funds PhotonDelta and 6G-Future Network Services. (Corresponding author: Eduardo Muller.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JLT.2026.3693164>.

Digital Object Identifier 10.1109/JLT.2026.3693164

Among OWC strategies, narrow-beam infrared transmission is particularly well-suited to indoor networking. A narrow beam allows multiple users in the same room to be served simultaneously with dedicated high-capacity links, avoiding shared-channel interference. The O-band (1260–1360 nm) was used as it offers the best eye safety at the 13 dBm power levels needed for indoor use [4], [5]. Although 850 nm sources are more commercially mature, their shorter wavelength poses greater eye-safety risks for user-facing indoor links. And the 1550 nm wavelength, despite its maturity in fiber-optic technology, suffers from degraded performance under higher attenuation conditions [4] and significant thermal limitations in VCSEL development due to the poor thermal conductivity of InP-based materials [6]. This combination enables multi-Gbps data rates per user while maintaining user privacy through spatial selectivity [3]. However, leveraging these advantages requires precise, rapid beam steering so that the narrow optical beam remains aligned with the receiver aperture, even as users move.

The considered indoor communication architecture is built on a fiber to the room (FTTR) infrastructure, as depicted in Fig. 1(a) [7]. Optical signals at user-specific wavelengths are multiplexed and delivered to a ceiling-mounted transmitter, where they are separated by a demultiplexer and detected by a photodiode array. The resulting electrical signals, from the photodiodes, are routed in real time by a field-programmable gate array (FPGA) to control a two-dimensional vertical-cavity surface-emitting laser (VCSEL) focal plane array (FPA). The FPGA can be controlled by a user localization technique to drive one or more corresponding VCSELs, like the one showed in [8]. A single shared optical lens projects the selected VCSEL output toward the intended user, enabling electronic beam steering without mechanical motion, as illustrated in Fig. 1(b).

In our previous work [9], we demonstrated the feasibility of this concept using a basic 2 × 2 VCSEL FPA. The present work advances the concept significantly by scaling to a 4 × 4 VCSEL array, enabling individual addressing of 16 emitters and simultaneous servicing of multiple users. We experimentally validate real-time beam steering with a steering range of 10° × 15°, achieving error-free transmission (bit error-rate (BER) < 10^{−9}) at 10 Gbps over indoor free-space distances up to 4 m. This represents, to the best of our knowledge, the first demonstration of a multi-user, electronically steered VCSEL focal plane array operating at multi-Gbps data rates for indoor OWC.

These results establish VCSEL FPAs as a compelling solution for next-generation indoor wireless networks, combining low

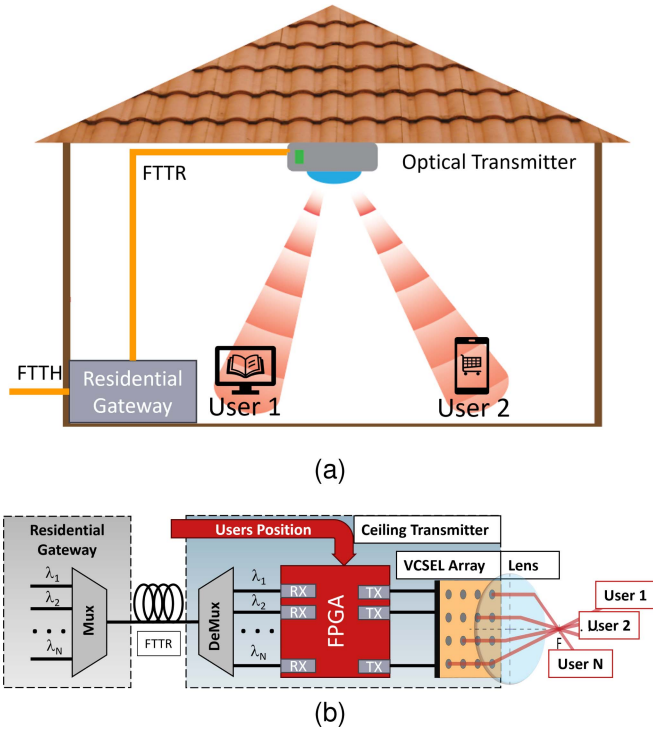


Fig. 1. Proposed indoor optical wireless communication system architecture. (a) FTTR infrastructure with narrow optical beam transmission for last-meter connectivity, (b) system topology illustrates data flow from the residential gateway to the ceiling-mounted transmitter unit, employing a VCSEL-FPA array for beam steering.

system complexity, fast electronic reconfiguration, high spatial precision, and support for multiple simultaneous users. The remainder of this paper is organized as follows: Section II reviews related beam-steering approaches. Section III describes the proposed system architecture and beam-steering concept. Section IV details the experimental setup. Section V presents the measurement results and discussion. Section VI concludes the work and outlines future research directions.

II. OPTICAL BEAM STEERING TECHNIQUES

A variety of optical beam-steering approaches have been investigated for indoor OWC systems. Their suitability depends on steering flexibility, response speed, system complexity, and compatibility with high-speed data transmission. This section reviews representative techniques and evaluates their applicability for indoor OWC.

Polarization gratings provide a beam-steering mechanism in which thin liquid-crystal layers generate switchable ± 1 diffraction orders through controlled polarization states [10]. Although stacked grating architectures offer low power consumption and compact integration, they demand stringent polarization management, which limits their suitability for fiber-fronthaul OWC systems. Furthermore, liquid-crystal response times remain comparatively slow, and both insertion loss and system complexity increase substantially when larger steering angles are realized through multi-grating cascades.

Arrayed waveguide grating (AWG) routers offer a passive wavelength-controlled steering solution. By tuning the emission wavelength, the output beam is directed to different fiber positions located at the transmitter's focal plane [3]. This architecture naturally supports multi-beam operation and can scale to large arrays. While AWG-based steering depends on tunable laser sources, careful design of beam divergence, aperture size, and wavelength spacing allows balancing coverage area and spatial resolution. Consequently, AWG systems are attractive for scenarios where the transmitter or receiver positions are approximately known, localization assistance is present and available wavelength range is sufficient large to cover possible user locations [11].

Metasurface-based beam steering employs sub-wavelength resonator arrays to impart spatially varying phase shifts [12]. These ultra-thin structures provide compact form factors and fine phase control. However, metasurfaces generally exhibit limited operational bandwidth and are often optimized for one-dimensional steering. Beam squint effects, arising from the frequency dependence of the engineered phase response, further constrain wide-angle operation. Additionally, each metasurface must be fabricated for a specific wavelength and steering configuration, limiting reconfigurability and scalability.

Optical phased arrays (OPAs) electronically control the emitted wavefront through integrated phase shifters [13]. They offer solid-state beam steering without mechanical movement and can be integrated in photonic platforms. Nevertheless, precise phase control across large 2D arrays remains challenging. Achieving two-dimensional steering often requires combining wavelength tuning with thermo-optic or electro-optic phase adjustment, increasing both control complexity and power consumption [14]. Fabrication tolerances strongly influence system stability and beam quality, especially as array sizes scale.

Spatial light modulators (SLMs) steer beams by modulating refractive index patterns across a 2D pixel matrix [15]. SLMs provide dynamic, fully programmable wavefront shaping and can generate multiple simultaneous beams. However, steering speed is limited by pixel refresh rates, and achievable resolution is constrained by pixel pitch. These factors limit system responsiveness for real-time user tracking in high-mobility indoor environments.

Compared to these techniques, VCSEL focal plane arrays provide a particularly favorable trade-off for high-speed indoor OWC. Beam steering is achieved by electronically selecting individual emitters, eliminating complex phase control or wavelength tuning mechanisms. These VCSELs can all work on nominally the same wavelength. This leads to inherently low system cost, fast response times, and straightforward scalability to support multi-user access. Moreover, the achievable scan range can be made large by appropriate optical design, enabling flexible link establishment across typical indoor line-of-sight distances.

III. SYSTEM DESIGN AND SETUP

To validate the proposed indoor OWC architecture, we developed a functional transmitter prototype comprising a 4×4

VCSEL FPA, an FPGA-based high-speed routing and control system, and a lens-based beam-steering optical stage. The transmitted signal is received by a high-bandwidth fiber photoreceiver used as an idealized reference receiver.

A. VCSEL Array

The 4×4 VCSEL array (1300 nm, Alight Technologies) was selected for its compact footprint and high-speed modulation performance while remaining within Class 1 eye-safety limits. Each VCSEL supports up to 10 Gbps non-return-to-zero on-off keying (NRZ-OOK) transmission, corresponding to an electrical 3 dB modulation bandwidth of approximately 8 GHz. After accounting for coupling and interposer reflections, each emitter delivers approximately 0 dBm optical power into free space.

The 16 VCSELs are arranged as four linear 1×4 subarrays with a $250 \mu\text{m}$ intra-subarray pitch and a $350 \mu\text{m}$ inter-subarray gap. The spacing accommodates the fan-out traces that interface with four Mellanox MTxV2804 VCSEL driver ICs. When all 16 VCSELs are driven simultaneously to reach the aggregate 160 Gbps capacity, thermal management becomes a challenge. Each VCSEL dissipates approximately 22 mW at the operating point, while each of the four quad VCSEL drivers dissipates approximately 800 mW at peak power, resulting in a total power budget of roughly 3.5 W. The thermal challenge comes primarily from the compact form factor and the driver ICs which require significantly more power than the emitters themselves. A compact aluminum heatsink of $12 \times 12 \times 18 \text{ mm}$, with a fan, can dissipate 3 to 5 W. In our prototype, wire-bonds presented a challenge for direct contact cooling, so external fans were used for heat dissipation. For a proper deployment, a dedicated heatsink or thermoelectric peltier cooler integrated into the transmitter module would be needed to ensure long-term stability. Total power dissipation can be further reduced by deactivating emitters and drivers that are not serving users, which is supported by the FPGA-based routing architecture.

Each VCSEL was biased at 3 mA with a modulation current of 7.8 mA, configured through the I^2C interface of the Mellanox drivers. These values were selected as they produced the best eye opening across all channels. The drivers support per-channel configuration of both bias and modulation current, allowing individual tuning per VCSEL to equalize optical output power or compensate for thermal variations between emitters at different positions. The per-emitter power dissipation is fixed at the optimal operating point and cannot be dynamically reduced without degrading signal quality. Reducing total system power consumption is achieved by selectively disabling unused VCSEL channels and their associated drivers when fewer users are active.

B. Steering Lenses

Beam steering is achieved by selecting which VCSEL within the array is active. The steering angle depends on the emitter's lateral displacement from the optical axis and the focal length of the collimation lens. This can be modeled through paraxial ray analyses. Considering the paraxial approximation for small angles where $\sin \alpha \approx \alpha$ and $\cos \alpha \approx 1$ [16], valid for steering

angles up to 10° [17]:

$$\begin{bmatrix} x_1 \\ \theta_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1/f & 1 \end{bmatrix} \begin{bmatrix} x_0 \\ \theta_0 \end{bmatrix} \quad (1)$$

$$\theta_1 = \theta_0 - \frac{x_0}{f} \quad (2)$$

where x_0 and x_1 represent the initial and final ray positions, θ_0 and θ_1 denote the initial and final ray angles, and f is the lens focal length.

The choice of steering lens is tightly coupled to the VCSEL array pitch since both parameters jointly determine the angular beam grid projected into the room. For a fixed emitter pitch, a shorter focal length increases the angular space between adjacent beams and widens the total field of view, but also increases beam divergence for a given VCSEL. Conversely, a longer focal length reduces the angular space and narrows the field of view, but produces tighter collimation and relaxes the divergence requirement since neighbouring beams are placed closer together relative to their spot size [7]. To accommodate a larger field of depth while maintaining collimation, the axial distance between the VCSEL array and the lens can also be made adjustable, allowing the beam position to be shifted for different target distances without changing the steering angles. In general, the lens selection defines a trade-off between coverage area, beam granularity, beam divergence, and inter-channel isolation, and can be adjusted independently of the VCSEL array to match different room geometries and user densities. We evaluated plano-convex spherical lenses with focal lengths of 4 mm, 6 mm, and 9 mm, with diameters of 2 mm, 3 mm, and 3 mm respectively. All lenses are NBK7 with a C (1050–1700 nm) anti-reflection coating. With (2) these lengths produce theoretical maximum steering angles of approximately 9.2° , 6.2° , and 4.1° respectively. The effective total field of view (FoV) is determined by the maximum separation of usable emitters within the FPA, meaning the VCSEL at the external border of the array.

The lens were verified using Zemax OpticStudio simulations to evaluate FoV coverage, beam spot size, and inter-beam interference for multi-user operation. The selected optical configuration achieves an experimental steering range of 10.7° horizontally and 15° vertically while preserving sufficient optical power and maintaining error-free transmission. The operational steering angles remain sufficiently small to satisfy the paraxial approximation, and the beam is confined to the central region of the lens, as the size of the array is smaller than the lens, avoiding spherical aberrations.

C. FPGA Control Logic

The FPGA is responsible for both routing user data to the appropriate VCSEL and managing beam steering commands. Fig. 4 shows the logic architecture.

A Vivado integrated bit-error rate tester (IBERT) pseudo-random binary sequence (PRBS) generator creates standardized 10 Gbps OOK test streams. The IBERT core must be externally bridged to the switch since the test stream generated is outputted directly to the high-speed transceivers, illustrated as the TX-RX connection on Fig. 4. Also, the IBERT core and

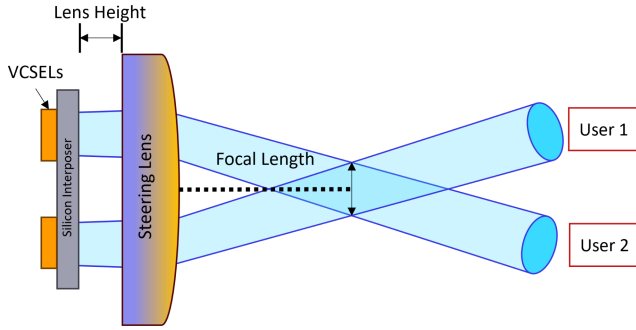


Fig. 2. VCSEL Focal Plane Array concept.

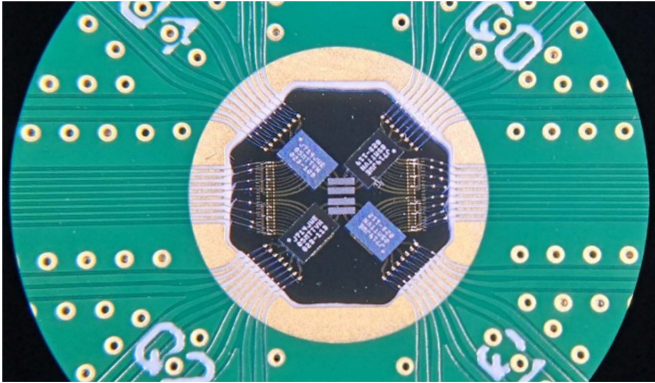


Fig. 3. The 4x4 VCSEL array and integrated driver assembly.

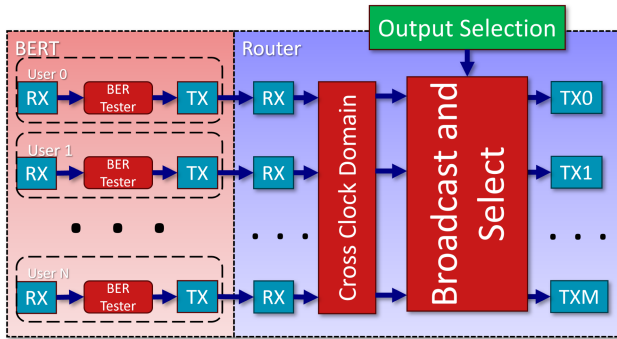


Fig. 4. FPGA routing and beam steering control architecture.

steering logic may operate on different internal clock domains, data is transferred across a cross-clock-domain first-in first-out (FIFO) to ensure safe synchronization. This design minimizes timing jitter and simplifies system timing by relying on a single external reference clock.

After synchronization, the streams enter a broadcast-and-select router, which maps each data stream to a chosen VCSEL channel based on steering control signals. This module enables real-time beam switching and can duplicate streams across adjacent emitters to support handover during user motion.

During prototyping, output addresses are manually added via a Virtual I/O (VIO) module. The control path is intentionally asynchronous and low-speed, enabling straightforward replacement with a future autonomous localization interface.

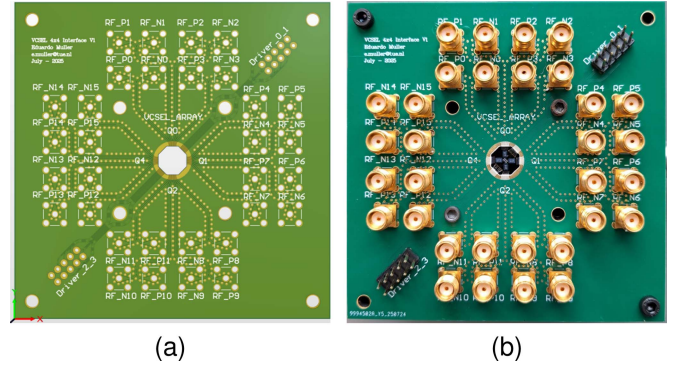


Fig. 5. Support PCB including the four drivers I2C channels and 16 high-speed differential SMA waveguides, (a) Gerber layout showing differential routing; (b) fully assembled board with drivers and VCSEL array.

The selected data signal is finally serialized through a Gigabit high-speed transceiver (GTH) and delivered to the corresponding VCSEL driver. Each optical channel was individually equalized using transceiver pre-emphasis tuning to ensure open data eyes at 10 Gbps.

D. High-Speed Support PCB

The support PCB electrically interfaces the FPGA transceivers and the VCSEL drivers. It is fabricated on Rogers 4350B (0.5 mm) to ensure stable dielectric performance at multi-GHz operation. Differential coplanar waveguides are routed to maintain 100 Ω impedance. Quarter-wave impedance transformers are incorporated at driver pad transitions to minimize signal reflections.

The board supports 16 individual differential connections, one for each VCSEL, along with four I²C pinout connectors for the VCSEL drivers to the FPGA. This design enables reliable high-speed operation while accommodating the dense interconnect requirements of the system. The final assembly with all the connectors installed can be seen on Fig. 5(b).

E. Design Rationale Summary

Overall, the system architecture reflects a co-optimization of optical hardware, high-speed electronics, and control logic to enable reliable multi-point beam steering at 10 Gbps line rates. The VCSEL array provides closely spaced, individually addressable emitters with nanoseconds switching to support rapid spatial handoff. The choice and configuration of steering lenses balance FoV and beam quality, allowing controlled angular coverage without degrading the modulation performance. The FPGA-based routing and synchronization architecture isolates the high-speed data path from the lower-speed steering control, simplifying scalability to more advanced tracking and multi-user scheduling algorithms. Finally, the high-frequency PCB design preserves signal integrity from transmitter to optical output, ensuring that performance is limited by the optical channel rather than electronic bottlenecks. Together, these design decisions provide a robust foundation for dynamic OWC links that are scalable to larger arrays, wider coverage, and fully autonomous multi-user operation.

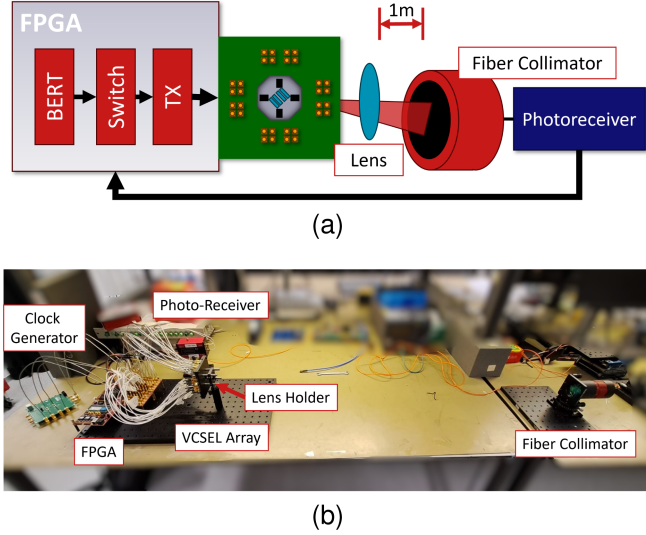


Fig. 6. Experimental setup: (a) system block diagram and (b) laboratory implementation.

IV. EXPERIMENTAL SETUP

To characterize the performance of the proposed OWC transmitter, we constructed the experimental setup illustrated in Fig. 6. The goal of this setup is to evaluate the wireless link while isolating the transmitter behavior from external penalties by ensuring a clean high-speed electrical drive and a high-bandwidth optical receiver.

The FPGA generates 10 Gbps NRZ-OOK PRBS data streams and performs beam steering using the broadcast-and-select routing logic described previously, as shown in Fig. 6(a). The selected data stream is delivered to the VCSEL driver through properly matched 100 Ω differential SMA interconnects and controlled-impedance PCB traces. To maintain signal integrity, all RF cables in the electrical path are length-matched and shielded to minimize skew and reduce crosstalk between channels.

The receiver configuration is designed to avoid imposing receiver-limited penalties and to accurately capture transmitted signal quality. The free-space optical beam is collected using a fiber collimator (Thorlabs C80FC, with a 42.5 mm aperture and 80 mm EFL) and coupled into a 50 μm multimode fiber. The coupled signal is detected using a 25 GHz TIA-integrated photoreceiver, providing sufficient bandwidth for error-free NRZ-OOK reception at 10 Gbps. The recovered data stream is then routed back to the FPGA, where real-time BER analysis is performed using the IBERT core.

Note that the receiver used in this experiment is deliberately configured with a narrow FoV and large electrical bandwidth. This ensures that the measured performance primarily reflects the transmitter's modulation quality and beam-steering accuracy, rather than limitations of the receiver optics or electronics. Although such a narrow-FoV receiver is unsuitable for mobile or wide-area coverage, it provides a controlled environment in which alignment, signal distortion, and BER can be directly attributed to the transmitter. In practical indoor deployments,

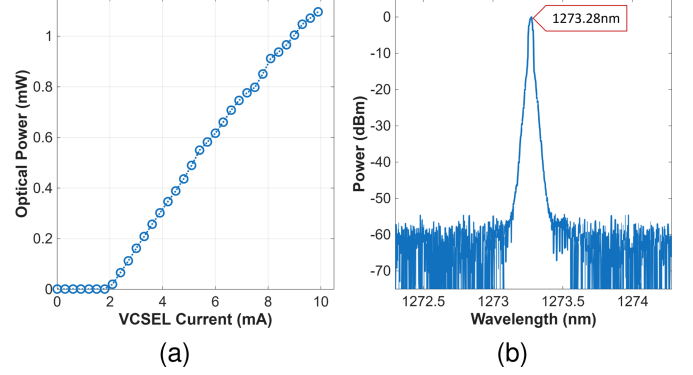


Fig. 7. Characterization of a single 1300 nm VCSEL from the array: (a) optical power versus Current, and (b) optical spectrum.

see [7] for example, a wide-FoV, lower-bandwidth receiver with spatial or angular diversity would be used to support mobility and relaxed alignment. Here, the narrow high-speed receiver serves mainly as a diagnostic tool to validate the feasibility of high-rate, beam-steered optical-wireless links.

V. RESULTS AND DISCUSSION

The performance of the proposed VCSEL FPA transmitter was evaluated in terms of link budget, signal integrity, and beam steering capability. Measurements were carried out initially at a 1 m link distance unless otherwise stated.

As an initial step, a single VCSEL from the array was characterized, with the L-I characteristics shown in Fig. 7(a). The current was measured from the VCSEL driver itself, stepped at 0.3 mA. This VCSEL exhibits a threshold current of approximately 1.8 mA. Above this threshold, the device enters the linear regime where the optical output power demonstrates an excellent linear dependence on the injection current. Each VCSEL can have approximately 14 mA driving current, corresponding to an optical output power of 1.6 dBm.

The optical spectrum in Fig. 7(b) indicates single-wavelength VCSEL operation at 1273.28 nm with an optical signal-to-noise ratio exceeding 55 dB. Thermal management of this prototype remains challenging due to the compact VCSEL layout and cabling, which hinder direct cooling. Although external fans were used for heat dissipation, stable temperature control could not be achieved.

To assess transmission reliability, the bit-error rate (BER) was measured as a function of received optical power for all sixteen VCSEL channels. The receiver configuration used for eye-diagram characterization was retained, and the input power was adjusted with a variable optical attenuator. Fig. 8 presents the BER versus received optical power for each of the four VCSEL quadrants, with all channels operating at 10 Gbps.

All sixteen channels achieve error-free operation below the maximum VCSEL output power, confirming full-array functionality at 10 Gbps. Most channels reach the Forward Error Correction (FEC) limit at sensitivities between -22 dBm and -23 dBm. A consistent intra-quadrant variation is observed: channels closer to the center of each driver region (e.g., Channels

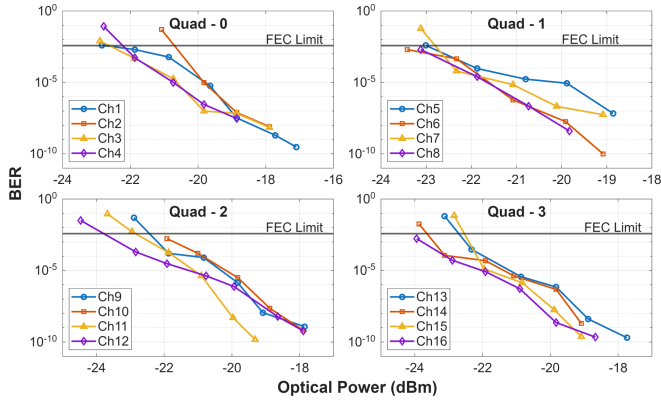


Fig. 8. BER performance vs. received optical power for the four VCSEL quadrants at 10 Gbps. The FEC threshold is indicated as a horizontal reference line. Measurements were taken at 1 m link distance.

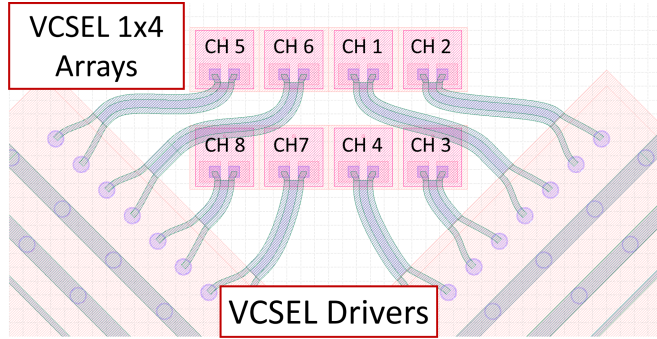


Fig. 9. Driver-to-VCSEL pad layout for the 4×4 VCSEL array. Trace lengths and routing asymmetries contribute to channel performance variation.

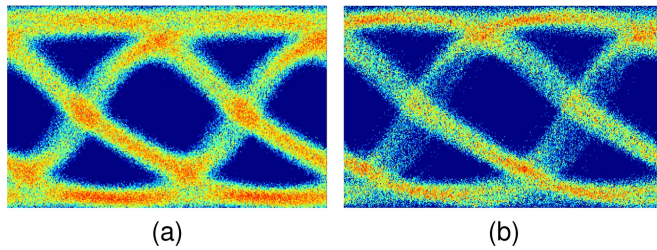


Fig. 10. Eye diagrams for the best-performing channel (a) Q3 CH4 and the worst-performing channel (b) Q0 CH2 at 10 Gbps over 1 m link distance.

2, 5, 10, and 13) require higher received power to reach the same BER. This trend correlates with the physical layout of the VCSEL-to-driver interconnects (Fig. 9), differences in trace geometry introduce variations in signal balance and bandwidth, leading to the observed spread. Nonetheless, even the worst-performing channels maintain ample margin for reliable data transmission.

Signal integrity was evaluated by comparing eye diagrams at the driver output and after optical reception. Fig. 10 shows representative eye diagrams for the best-performing (Q3 CH4) and worst-performing (Q0 CH2) channels.

The best-performing channel exhibits a wide eye opening with low amplitude noise and minimal timing jitter, indicating high

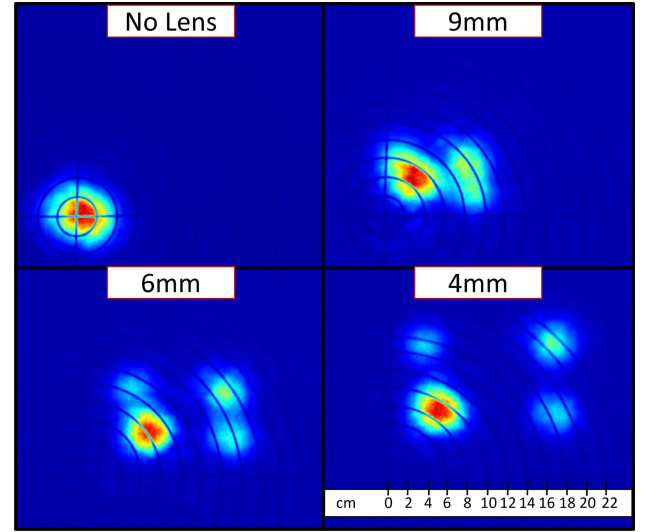


Fig. 11. Measured angular steering range for the focal lengths of 4 mm, 6 mm, and 9 mm, with four beams active at the same time (the rings are spaced 2 cm apart).

signal fidelity. The worst-performing channel shows increased jitter and amplitude fluctuations, attributed to differential trace imbalance in the electrical feed lines. Nevertheless, the eye remains sufficiently open to support error-free detection, consistent with the BER results.

Fig. 11 presents the measured beam steering patterns with four VCSELs from a single quadrant operating simultaneously, demonstrating both the multi-beam capability and spatial isolation of our system. Each visible spot represents an active VCSEL beam, with the four beams corresponding to the four emitters within one quadrant of the 4×4 array. Since a single quadrant contains all representative VCSEL positions (center, sides, and corner positions) and the array exhibits four-fold rotational symmetry, displaying one quadrant is sufficient to characterize the complete system behaviour while avoiding visual clutter from overlapping beam patterns. The measurements were conducted at 1 m distance for three different lens configurations (4 mm, 6 mm, and 9 mm focal lengths). The spatial separation between adjacent beam spots is clearly demonstrated with the 6 mm and 4 mm lenses, showing no optical overlap between simultaneously active channels and confirming interference-free multi-user operation. The beam patterns maintain distinct, well-isolated spots across all lens configurations, with the 4 mm lens providing the widest angular coverage at 9.2° while the 6 mm and 9 mm lenses reduce the range to 6.2° and 4.1° , respectively. All beams exhibit approximately 3.1° divergence, corresponding to a beam diameter of 55 mm at the measurement plane. Although the beams exhibit a circular profile, integrated micro-lenses may be used to further collimate the output and reduce divergence [18].

While Fig. 11 shows decreased light intensity for beams at larger steering angles with the 4 mm focal length lens, this intensity variation does not compromise link performance. The VCSELs are positioned at the focal plane of the steering lens to produce a nominally collimated output beam. The measured

full-angle divergence of approximately 3.1° , corresponding to a beam diameter of 55 mm at the 1 m measurement plane, arises primarily from the finite emitting aperture of the VCSEL and diffraction effects rather than defocus or misalignment. Because the beam is nominally collimated the received optical power does not degrade significantly with distance within the designed operating range. At the maximum tested distance of 4 m the received optical power remained approximately -16 dBm for all beam positions. As shown in Fig. 8 this power level is still error-free while providing a 6 to 7 dB margin above the FEC threshold. Consequently per-user power control is not required at indoor distances up to 4 m. It is important to note that the power variation observed between channels in Fig. 8 is primarily attributed to VCSEL placement tolerances on the interposer rather than beam steering effects with only 1.2 dB maximum difference between emitters. A shorter focal length inherently increases beam divergence for a given VCSEL aperture. To extend the operational range or reduce divergence, a longer focal length lens can be selected at the cost of a narrower field of view, or the axial VCSEL to lens distance can be made adjustable to reposition the beam waist for different target distances.

Importantly, the spatial separation of the steered beams is sufficient to prevent inter-channel optical overlap, preserving channel isolation and enabling potential multi-user operation. No measurable BER degradation was observed during steering, indicating negligible beam distortion from the optics.

To assess applicability in realistic indoor environments, the link distance was increased to 4 m. Error-free operation at 10 Gbps was maintained across this range, confirming that the system supports room-scale coverage without performance loss.

Overall, these results demonstrate that the proposed 4×4 VCSEL focal-plane-array transmitter delivers stable, high-speed wireless optical links with electronically selectable beam directions, without mechanical actuation or complex beamforming. The system maintains error-free 10 Gbps transmission across all channels, exhibits predictable and well-isolated beam-steering behavior, and supports room-scale communication distances. These findings validate the FPA-based approach as a practical and scalable solution for future multi-user indoor OWC systems requiring 25 GHz bandwidth, spatial selectivity, and link privacy.

VI. CONCLUSIONS AND FUTURE WORKS

This paper presented the design and experimental validation of a compact, solid-state beam-steering optical wireless communication transmitter. The transmitter is based on a 4×4 VCSEL focal plane array operating at 1300 nm, which enables high-speed transmission while remaining compliant with indoor eye-safety requirements. Error-free operation was demonstrated across all 16 channels at 10 Gbps per channel, corresponding to an aggregate capacity of 160 Gbps. This performance was maintained over wireless link distances up to 4 m, representing a realistic operating range for indoor access point applications.

A key feature of the system is its tunable steering capability, achieved simply by selecting commercially available lenses with different focal lengths. Steering angles from 4.1° to 9.2° were demonstrated, enabling the transmitter's spatial coverage to

be adapted to diverse room geometries and user distributions. The current configuration produced beams with a maximum divergence of 3.1° , concentrating optical power at the receiver and providing inherent spatial isolation between simultaneously active links. Lower divergence can be achieved through lens optimization, which is particularly beneficial in multi-user environments where minimizing inter-user interference and ensuring link privacy are essential. Moreover, the beam spot size can be further adjusted by varying the lens-to-array separation, offering additional system-level flexibility without the need for mechanical beam steering.

Overall, this work validates that a compact 1×1.4 mm² VCSEL focal plane array transmitter, that can deliver high throughput, spatially selective optical wireless links suitable for next-generation indoor networks. Its combination of multi-gigabit performance, electronic beam steering, small footprint, and scalable architecture makes it well-suited for deployment in ceiling-mounted network access points as well as integration in space- and power-constrained user devices. This approach represents a promising pathway toward high-capacity, interference-aware, and privacy-preserving indoor optical wireless communication systems.

Looking ahead, several avenues can further advance VCSEL-FPA-based systems. Increasing array size would enable finer beam granularity, higher user capacity, and more flexible spatial multiplexing. Integrating micro-optic or metasurface beamforming elements could provide continuous (rather than discrete) steering while maintaining a solid-state design. Enhancements in VCSEL modulation bandwidth and driver equalization may support per-channel rates beyond 25 Gbps, enabling terabit-class aggregate capacities in similar form factors.

ACKNOWLEDGMENT

The authors would like to thank Alight Technologies from Denmark for providing us with the 1300 nm VCSEL array samples.

REFERENCES

- [1] Cisco, "Cisco annual internet report (2018–2023) white paper," [Online]. Accessed: Apr. 25, 2024. [Online]. Available: <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>
- [2] International Telecommunication Union, "Radio Regulations - Articles - Edition of 2020," Geneva, Switzerland: International Telecommunication Union, 2020. [Online]. Available: <http://handle.itu.int/11.1002/pub/814b0c44-en>
- [3] T. Koonen, "Indoor optical wireless systems: Technology, trends, and applications," *J. Lightw. Technol.*, vol. 36, no. 8, pp. 1459–1467, Apr. 2018.
- [4] Shaina and A. Gupta, "Comparative analysis of free space optical communication system for various optical transmission windows under adverse weather conditions," *Procedia Comput. Sci.*, vol. 89, pp. 99–106, 2016.
- [5] International Electrotechnical Commission (IEC), "Safety of laser products – Part 1: Equipment classification and requirements," IEC Std. 60825-1, 3.0 Ed., Geneva, Switzerland, May 2014.
- [6] A. Babichev et al., "Impact of device topology on the performance of high-speed 1550 nm wafer-fused vcsels," *Photonics*, vol. 10, no. 6, Jun. 2023, Art. no. 660.
- [7] T. Koonen, *Short-Reach Optical Wireless Communication: By Directed Narrow Beams*. Eindhoven, The Netherlands: IEEE, May 2025.

- [8] N. Q. Pham, K. Mekonnen, E. Tangdiongga, A. Mefleh, and T. Koonen, "User localization and upstream signaling for beam-steered infrared light communication system," *IEEE Photon. Technol. Lett.*, vol. 33, no. 11, pp. 545–548, Jun. 2021.
- [9] E. Muller, Y. Song, T. Koonen, and E. Tangdiongga, "Focal plane array using vcsels for beam steering in high-speed indoor optical wireless communication," in *Proc. 51th Eur. Conf. Opt. Commun.* Copenhagen: IEEE, Sep. 2025, pp. 388–391.
- [10] C. Hoy, J. Stockley, J. Shane, K. Kluttz, D. McKnight, and S. Serati, "Non-mechanical beam steering with polarization gratings: A review," *Crystals*, vol. 11, no. 4, Mar. 2021, Art. no. 361.
- [11] T. Koonen, K. Mekonnen, Z. Cao, F. Huijskens, N. Q. Pham, and E. Tangdiongga, "Ultra-high-capacity wireless communication by means of steered narrow optical beams," *Philos. Trans. Roy. Soc. A: Mathematical, Phys. Eng. Sci.*, vol. 378, no. 2169, Apr. 2020, Art. no. 20190192.
- [12] G. K. Shirmanesh, R. Sokhoyan, P. C. Wu, and H. A. Atwater, "Electro-optically tunable multifunctional metasurfaces," *ACS Nano*, vol. 14, no. 6, pp. 6912–6920, Jun. 2020.
- [13] N. Dostart et al., "Serpentine optical phased arrays for scalable integrated photonic lidar beam steering," *Optica*, vol. 7, no. 6, Jun. 2020, Art. no. 726.
- [14] J. K. Doylend, M. J. R. Heck, J. T. Bovington, J. D. Peters, L. A. Coldren, and J. E. Bowers, "Two-dimensional free-space beam steering with an optical phased array on silicon-on-insulator," *Opt. Exp.*, vol. 19, no. 22, Oct. 2011, Art. no. 21595.
- [15] A. Gomez et al., "Beyond 100-Gb/s indoor wide field-of-view optical wireless communications," *IEEE Photon. Technol. Lett.*, vol. 27, no. 4, pp. 367–370, Feb. 2015.
- [16] F. L. Pedrotti, L. M. Pedrotti, and L. S. Pedrotti, *Matrix Methods in Paraxial Optics*, 3rd ed. Cambridge, MA, USA: Cambridge Univ. Press, Dec. 2017, pp. 396–418. [Online]. Available: <https://www.cambridge.org/core/product/identifier/9781108552493/type/book>
- [17] M. Wolny, E. Muller, and E. Tangdiongga, "Transmitters and receivers for high capacity indoor optical wireless communication," *Telecom*, vol. 6, no. 2, Apr. 2025, Art. no. 26.
- [18] Y. Song, *Compact Optical Fiber and Wireless Interconnects: Micro-lens on Inerposer*. Eindhoven, Netherlands: Technische Universiteit Eindhoven, Dec. 2023, pp. 61–83. [Online]. Available: https://pure.tue.nl/ws/portalfiles/portal/313511831/20231219_Song_Y._hf.pdf